

M08 Module Review

Files, APIs & Data Ingestion | Review after attempt

Review rule

Open only after a genuine attempt. A review is diagnostic material, not the first way to solve the task.

M08-L01 - Ingestion architecture: source -> landing -> staging

Strong evidence: You can trace one source record through raw, staged/rejected and checkpoint outcomes.

Repair focus: Writing directly into staging/serving and losing the original source evidence.

M08-L02 - Defensive CSV ingestion

Strong evidence: Every input row has an explainable staged/quarantined outcome.

Repair focus: Dropping bad rows silently and reporting a clean batch count.

M08-L03 - JSON and nested-record ingestion

Strong evidence: You can state source object grain and staging row grain and explain the row-count change.

Repair focus: Flattening nested arrays without preserving parent keys, creating ambiguous rows.

M08-L04 - Parquet ingestion and metadata inspection

Strong evidence: You can state what metadata proves without loading the whole dataset.

Repair focus: Converting CSV to Parquet but preserving wrong inferred types and calling it solved.

M08-L05 - REST API contracts, status codes and timeouts

Strong evidence: Transport success and data-contract success are checked separately.

Repair focus: No timeout, or treating any JSON body as success even when status is an error.

M08-L06 - API authentication, pagination and rate limits

Strong evidence: Page counts and final record count prove pagination completeness.

Repair focus: Embedding a real token in source or stopping after page 1 because the response looks valid.

M08-L07 - Retries, backoff, jitter and transient failures

Strong evidence: Attempt log shows bounded retries and final outcome.

Repair focus: Infinite retry loops or retrying authentication failures that require configuration repair.

M08-L08 - Database extraction and safe query boundaries

Strong evidence: The query is read-only, bounded and deterministic under tied timestamps.

Repair focus: Timestamp-only incremental query that can skip rows sharing the same update time.

M08-L09 - Historical batch landing and raw metadata

Strong evidence: A raw file can be matched unambiguously to a run and checksum.

Repair focus: Overwriting old raw batches and later being unable to reproduce a historical run.

M08-L10 - Incremental loads and high-water marks

Strong evidence: You can prove why every row $\leq$ checkpoint has been fully handled.

Repair focus: Updating state after page 1 and then failing on page 2, causing missed records next run.

M08-L11 - CDC concepts and change-event semantics

Strong evidence: You can explain insert/update/delete events, replay and the ordering key.

Repair focus: Applying change events twice or out of order without an idempotent key/position rule.

M08-L12 - Schema drift, compatibility, time zones and quarantine

Strong evidence: No incompatible schema change reaches staging silently.

Repair focus: Auto-accepting renamed/retargeted fields and producing plausible values under the wrong contract.

M08-L13 - Checkpoint/state management and replay safety

Strong evidence: Replay/backfill behavior is explicit and state transition evidence is saved.

Repair focus: Replaying a historical run and unintentionally moving the production checkpoint backward/forward.

M08-L14 - Idempotent landing and content identity

Strong evidence: You can explain source-object identity vs business-record identity.

Repair focus: Skipping a changed file because its filename was seen before.

M08-L15 - Late-arriving data, event time and processing time

Strong evidence: You can state which timestamp drives business ordering, checkpointing and operational monitoring.

Repair focus: Sorting by ingestion time when the business logic requires event chronology.

M08-L16 - Ingestion observability: counts, latency and run metadata

Strong evidence: A reviewer can reconstruct the run outcome from metadata without reading the source code.

Repair focus: Logging only "completed successfully" with no completeness counters.

M08-L17 - Ingestion integration, reconciliation and backfill proof

Strong evidence: The end-to-end evidence proves completeness, quarantine, state and rerun behavior.

Repair focus: Passing each small lab but failing end-to-end completeness or recovery.